VOLTpark

- **Theme-** SDG 11 – Sustainable Cities and Communities, SDG 9 – Industry, Innovation and Infrastructure, SDG 13 – Climate Action.
- **Category-** Software
- **Group No - 17**
- **Group Members-** Neev Rambhia, Raj Dabholkar, Naman Shah
- **Guide-** Prof. Kanchan Wankhede

Introduction

Background of the Topic

- Rapid urbanization has increased vehicle density in cities.
- Finding parking spaces and EV charging stations has become time-consuming.
- Lack of real-time information leads to traffic congestion.
- Growth of electric vehicles requires accessible charging infrastructure.

Importance of the Problem

- Vehicles waste significant fuel while searching for parking.
- Increased congestion leads to higher carbon emissions.
- Poor parking management affects urban mobility efficiency.
- Smart-city initiatives demand digital infrastructure solutions.

Current Challenges

- No centralized platform for parking and EV charging availability.
- Lack of real-time slot visibility.
- Inefficient manual management of parking spaces.
- Limited digital integration in urban infrastructure systems.

Introduction

Observations That Motivated This Project

- Frequent traffic congestion due to vehicles searching for parking.
- Rising fuel costs and environmental concerns.
- Increasing adoption of electric vehicles but limited charging visibility.
- Need for smarter urban mobility systems.

SDG Mapping

- SDG 11 – Sustainable Cities and Communities: Improve urban mobility and infrastructure efficiency.
- SDG 13 – Climate Action: Reduce emissions caused by unnecessary vehicle movement.
- SDG 9 – Industry, Innovation & Infrastructure: Promote digital solutions for smart infrastructure.

Introduction

Project Objectives

- Provide real-time visibility of parking and EV charging availability.
- Enable advance booking of slots to reduce congestion.
- Digitize parking and charging infrastructure management.
- Improve resource utilization and reduce fuel wastage.
- Support sustainable and smart-city mobility solutions.

SDG Alignment

- Contributes to SDG 11 by improving urban mobility efficiency.
- Supports SDG 13 by lowering emissions and fuel waste.
- Promotes SDG 9 through digital innovation in infrastructure.

Literature Survey

Real-World Platforms

1. Park+ (India)

- Provides parking discovery and FASTag integration
- Offers digital parking booking in select areas

2. Parkopedia (Global)

- Database of parking locations worldwide
- Provides availability and pricing information

3. ChargeMap (Europe)

- EV charging station locator
- Shows charger types and availability

4. Google Maps

- Shows parking areas and EV charging stations.
- Limited real-time slot availability.

Literature Survey

Survey of Existing Platforms

- ParkWhiz / SpotHero (USA): Pre-booking with fixed lots — no real-time slot visualization [1]
- EasyPark (Europe): Sensor-based parking — requires expensive IoT hardware [2]
- Google Maps Parking: Suggests lots but no booking or slot-level management
- PlugShare / ChargePoint: EV charging locators only — no integrated parking [3]

Research Gaps Identified

- No unified platform combining parking + EV charging + booking + payment
- Most systems rely on hardware sensors; no purely software-driven solutions
- Owner-side management and admin oversight absent in consumer apps

Literature Survey

Survey of Methodologies / Techniques Used

Common Techniques in Smart Parking Research

- IoT-based sensor systems for slot detection
- RFID and Automatic Number Plate Recognition (ANPR)
- GPS and map-based location tracking
- Cloud-based centralized databases
- Mobile/web-based booking interfaces

Data Usage in Existing Systems

- Sensor-generated real-time availability data
- Historical booking patterns
- User geolocation data
- Payment transaction logs

Limitations

Limitations Identified from Literature Survey

- No single platform unifies parking discovery, EV charging, and digital booking
- Heavy dependence on IoT hardware sensors raises deployment cost and complexity
- Real-time visual slot layout is unavailable to users before arrival
- Multi-vehicle type support (cars vs. bikes) with separate pricing is missing
- Owner-facing management tools are absent in all surveyed consumer platforms
- No unified role-based access control (user / owner / admin) in open platforms
- Payment integration is absent or relies on poor third-party redirect UX.

Problem statement and Objectives

Problem Statement

- Urban drivers waste significant time and fuel locating parking or EV charging due to the absence of a unified real-time digital platform covering discovery, booking, and payment

.

Objectives

- Connect users with parking space and EV charging station owners via a web platform
- Enable real-time slot availability with an interactive visual parking layout
- Support separate slot management for cars and bikes with independent pricing
- Integrate Google Maps for location-based search of nearby facilities
- Process secure online payments via Razorpay with barcode booking confirmation
- Deliver dedicated dashboards for users, owners, and administrators (RBAC)

Proposed Methodology

System Architecture (MERN-Inspired)

- Frontend: React.js — component-based UI with dynamic slot visualization
- Backend: Supabase — PostgreSQL, REST API, real-time subscriptions, and Auth
- Payment: Razorpay Gateway — secure in-platform transaction processing
- Maps: Google Maps JavaScript API — location-based facility discovery
- Deployment: Vercel — continuous deployment with cloud hosting

Core Modules

- **User Module:** Search, filter, view availability, book, and pay for slots
- **Owner Module:** Register properties, manage slots, define pricing per vehicle type
- **Admin Module:** Approve owners, monitor activity, manage all listings
- **Real-Time Module:** Live slot updates via Supabase subscriptions on each booking

Proposed Methodology

Software Stack

- Frontend: React.js, HTML5, CSS3, JavaScript ES6+
- Backend-as-a-Service: Supabase (PostgreSQL, Auth, Realtime, Storage)
- Payment Gateway: Razorpay API
- Maps: Google Maps JavaScript API and Places API
- Deployment: Vercel (CI/CD pipeline with cloud hosting)

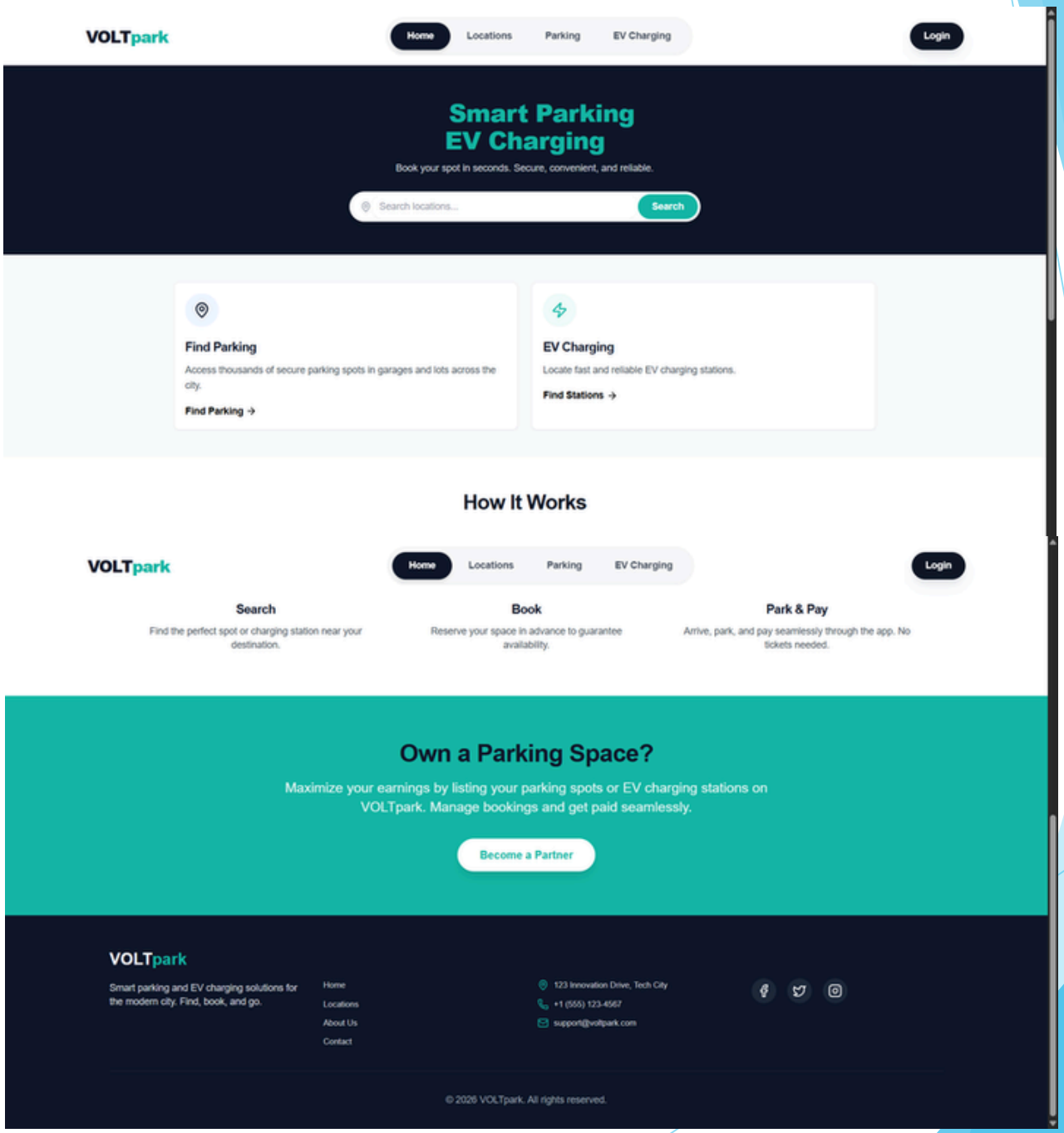
Development Environment

- IDE: Visual Studio Code | Version Control: Git & GitHub
- OS: Windows 10/11 | Testing: Chrome, Firefox, Edge

Evaluation Parameters

- Real-time slot update latency, booking success rate, page load time, payment success rate

Implementation



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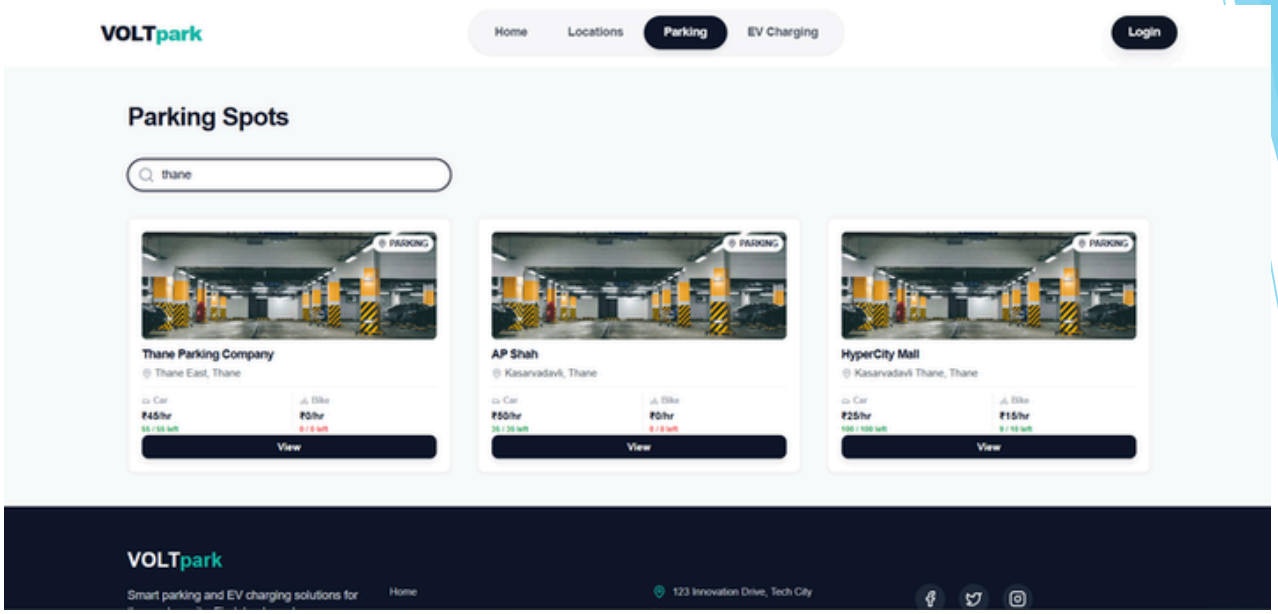

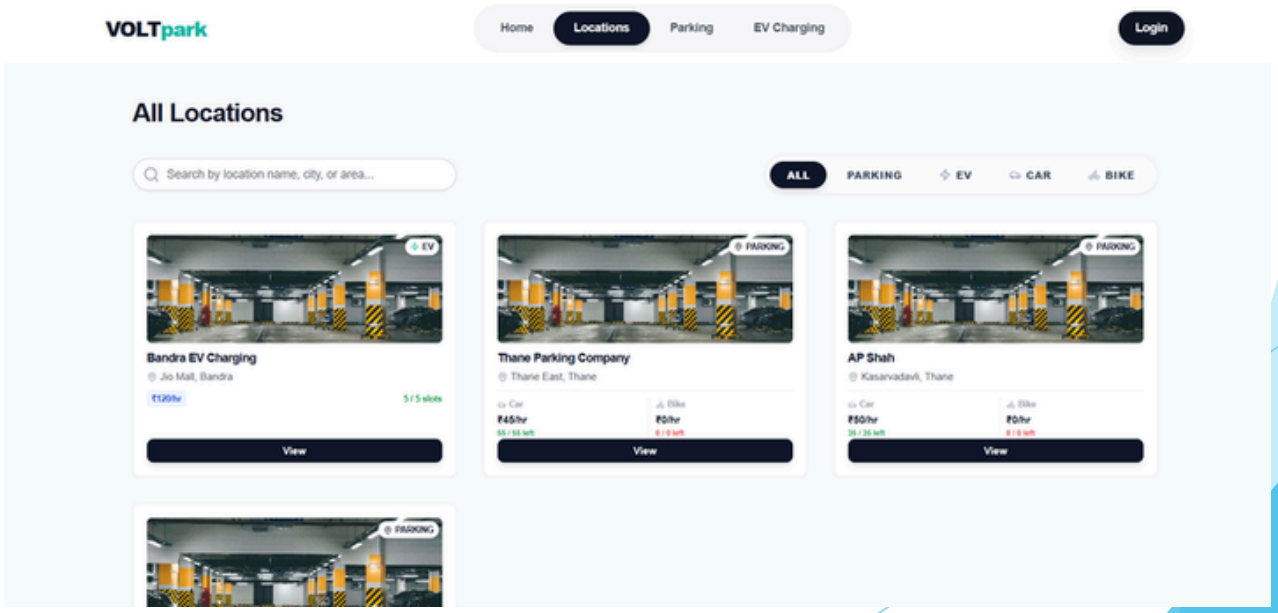

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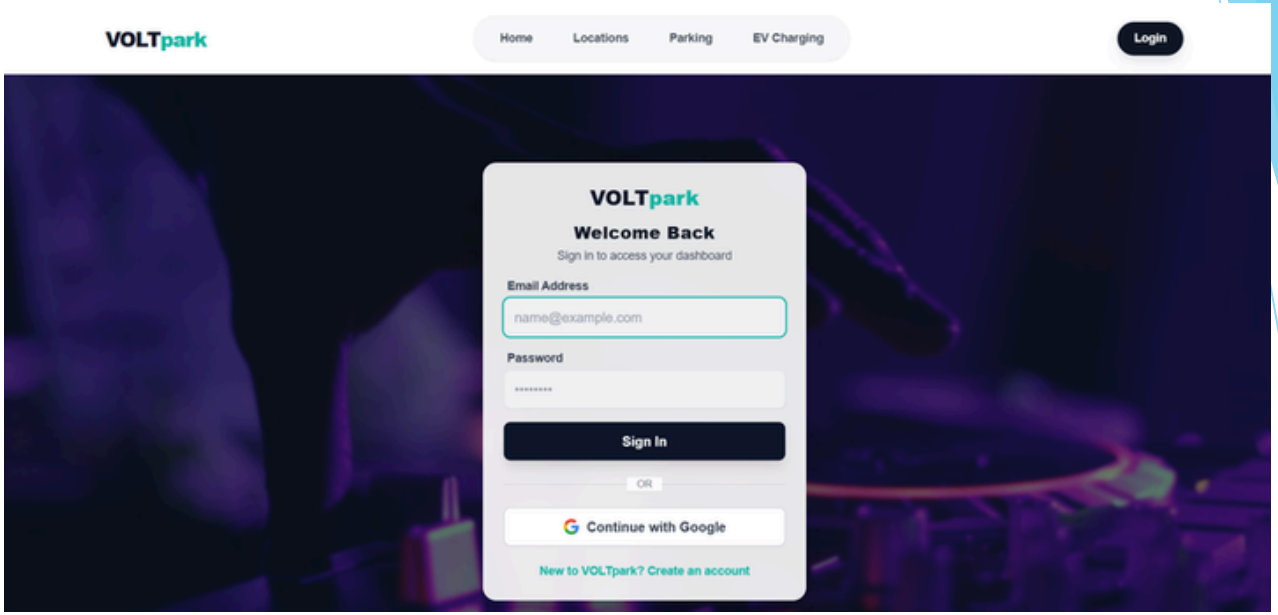

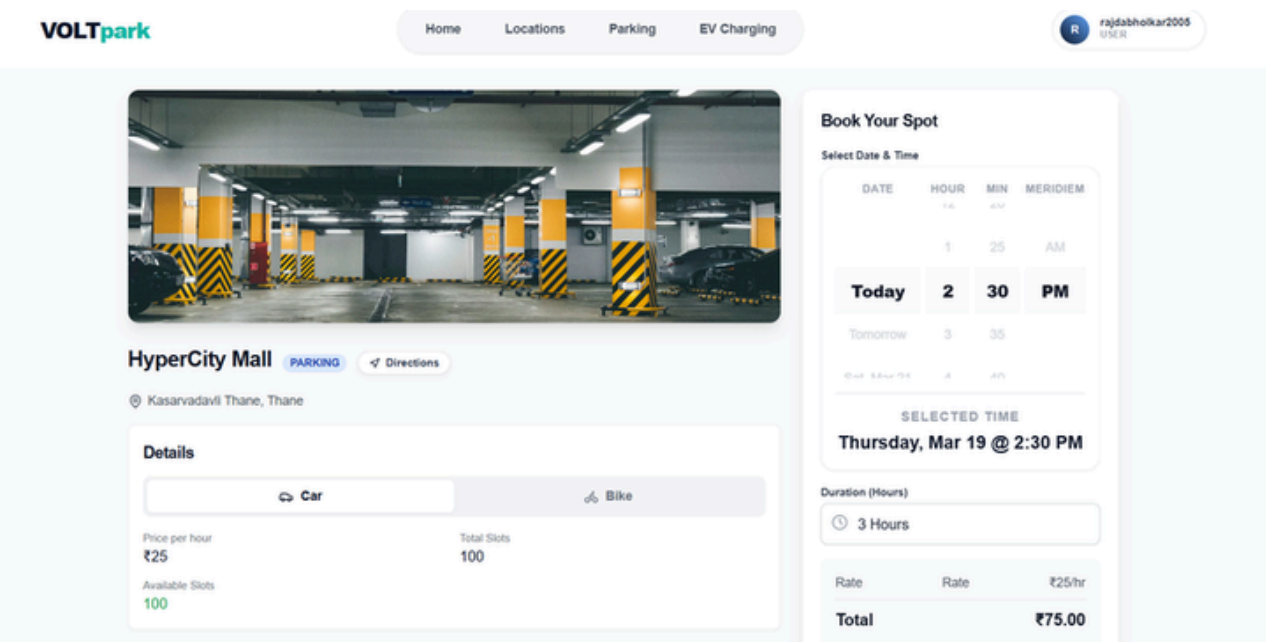

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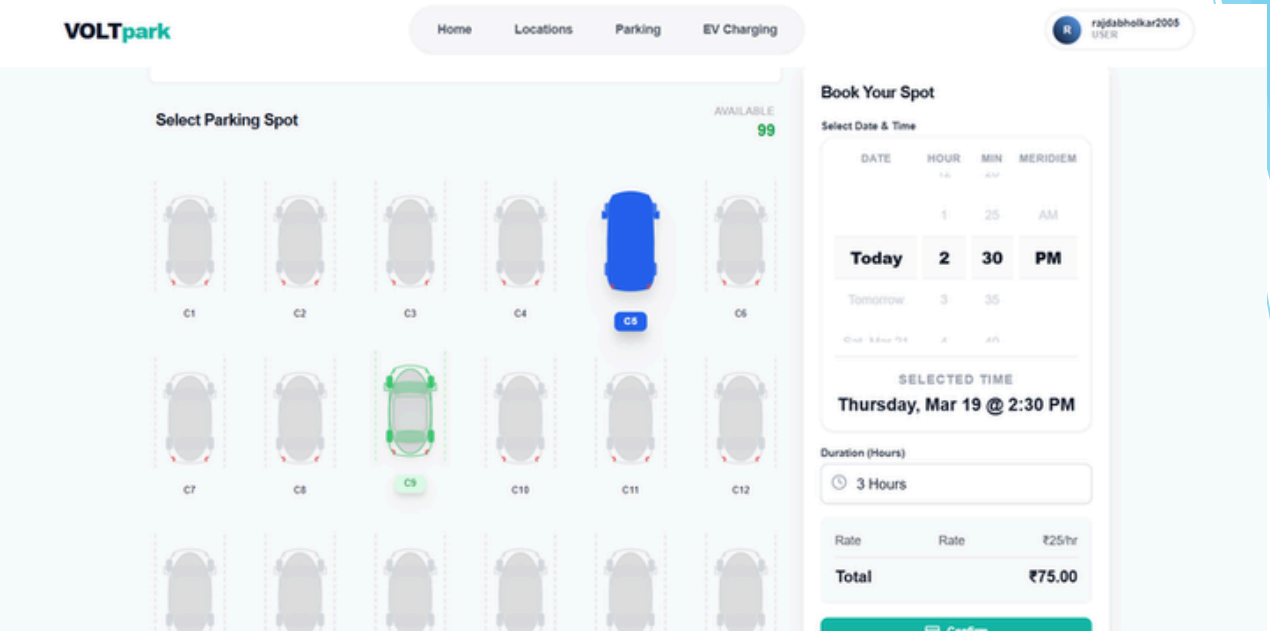

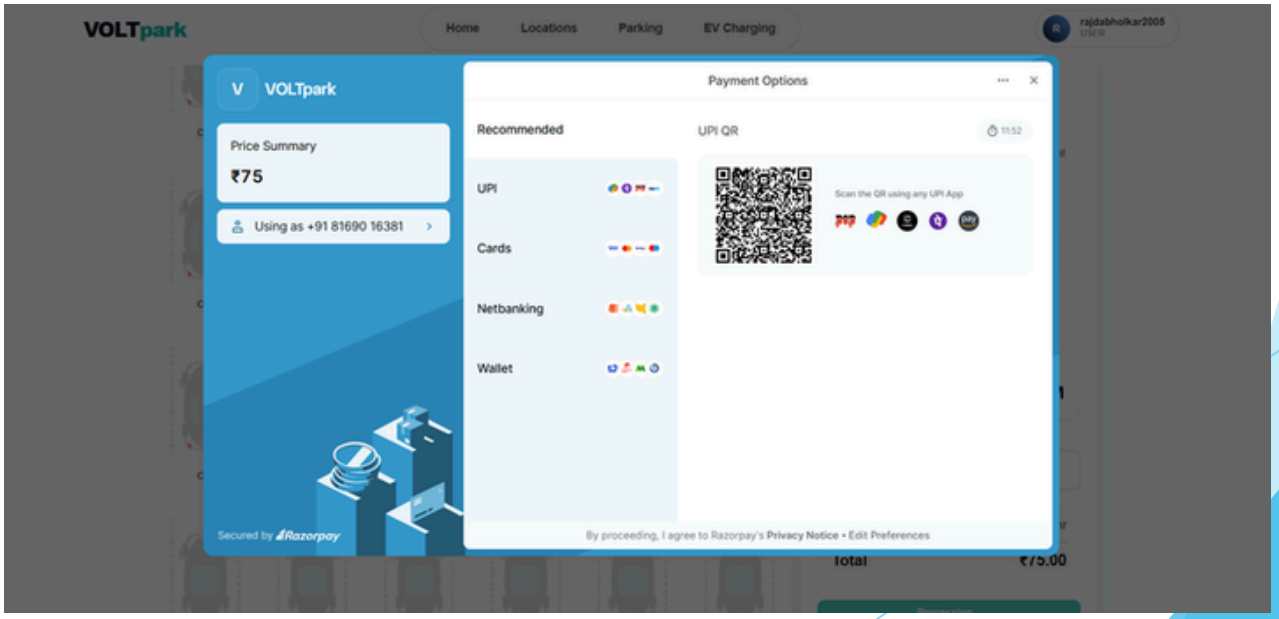

Results & Analysis

Platform Outcomes:

- Deployed on Vercel with stable uptime and real-time booking capability
- Slot updates reflect in under 1 second via Supabase live subscriptions
- All three role dashboards function independently with correct access control
- Razorpay payment integration: 100% success rate on sandbox transactions
- Google Maps correctly renders nearby facilities on city or location-based search
- Comparison with Existing Systems
- Uniquely combines parking + EV charging + booking + payment in one platform
- No IoT hardware required — fully software-driven slot management
- Scalable architecture supports new cities, owners, and vehicle types

Conclusion and Future work

Conclusion:

- VOLTpark digitizes parking and EV charging management for urban environments
- Delivers a scalable, role-based web platform with real-time availability and payments
- Mapped to SDG 9, 11, and 13 — supporting smart infrastructure and EV adoption
- Objectives Achieved
- Real-time slot visualization, multi-vehicle support, Google Maps, Razorpay payments
- Fully functional user, owner, and admin dashboards with role-based access control

Future Scope:

- IoT sensor integration for automated physical slot status detection
- Mobile application (Android/iOS) with push notifications
- AI-based demand prediction and dynamic pricing engine
- EV charging session monitoring with real-time energy usage tracking

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Thank You...!!